

Lending Club Case Study

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The problem

- ▶ Lending Club wants to understand how **consumer attributes** and **loan attributes** influence the tendency of default. The company can utilize this knowledge for its portfolio and risk assessment.
- ▶ As a Data Analyst working for Lending Club we analyze the dataset containing information about past loan applicants using EDA to understand how consumer attributes and loan attributes influence the tendency of default

Context

Problem statement

Analysis Approach

Clean Data

- Drop columns with null values or single category values
- Convert Values to proper data types

Univariate Analysis

- Check distributions and frequencies of various numerical and categorical variables
- Create derived variables

Segmented Univariate Analysis

- Analyze variables against segments of other variables
- Create derived variables

Bivariate Analysis

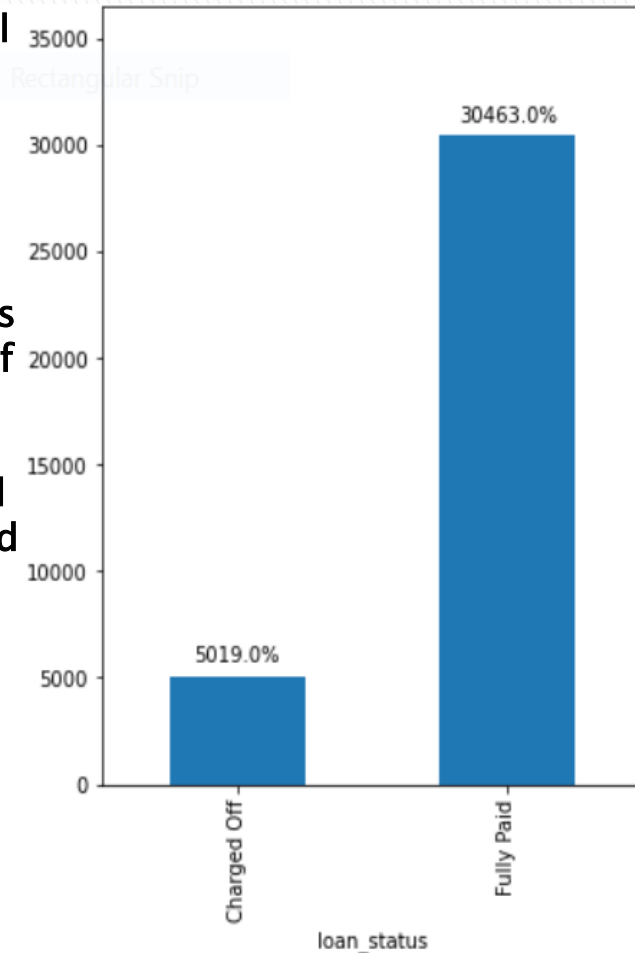
- Do correlation Analysis, Check how two variables affect each other or a third variable

Summarize Results

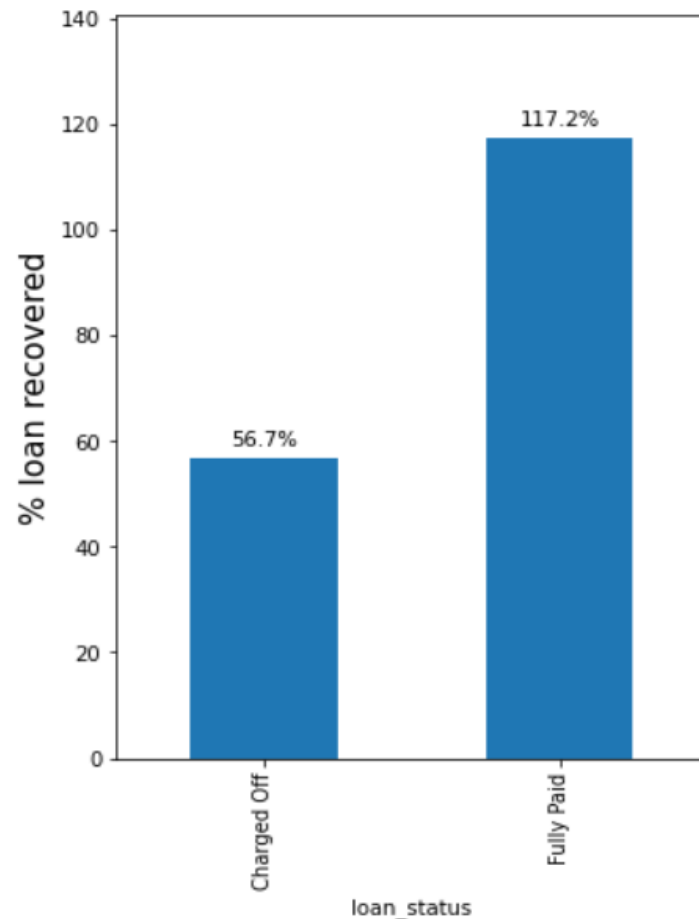
- Present Results

Analysis-Overall Loan Status

Total Loans

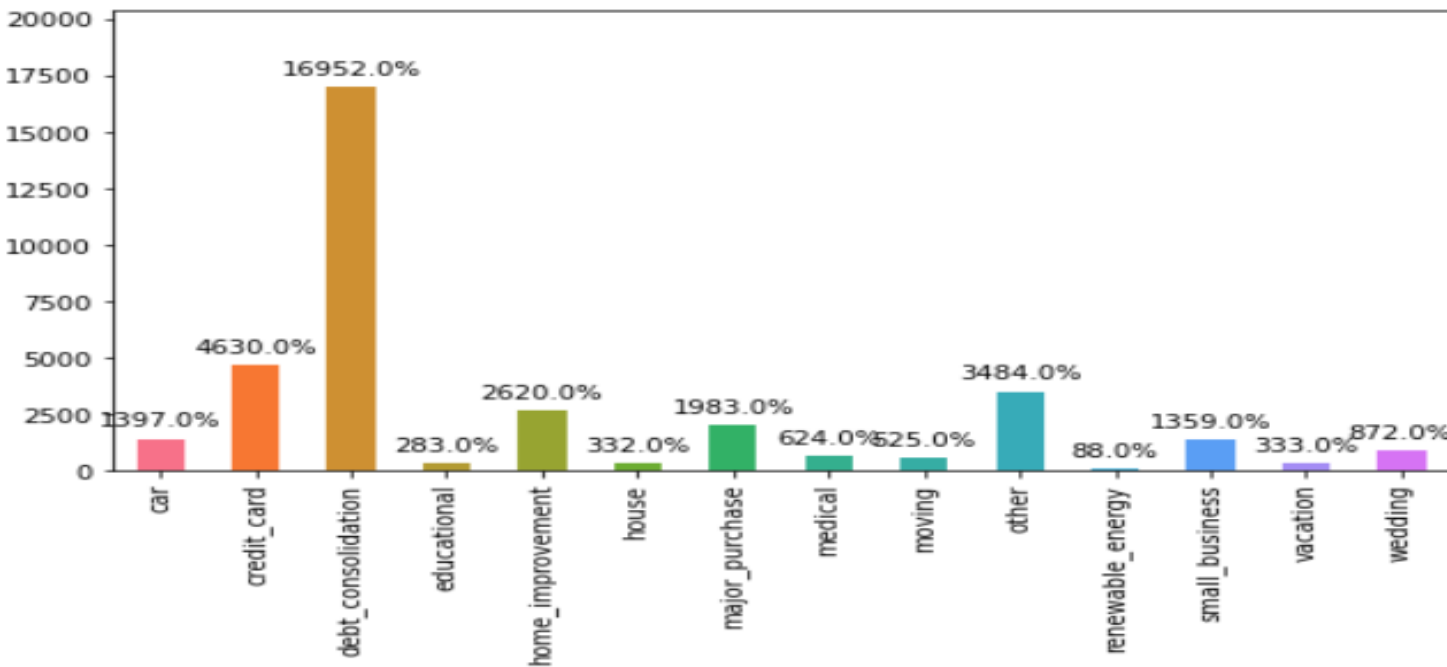


Total Money Earned

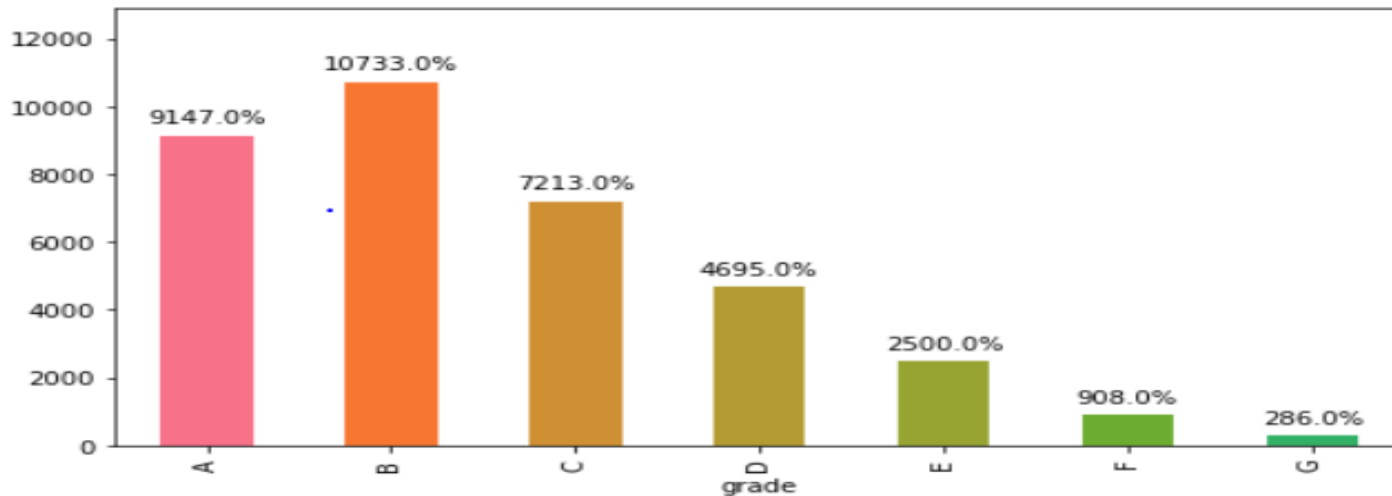


Lending Club only recover 56.7 % of the loan amount when the loan defaults

Analysis –Understanding Loans

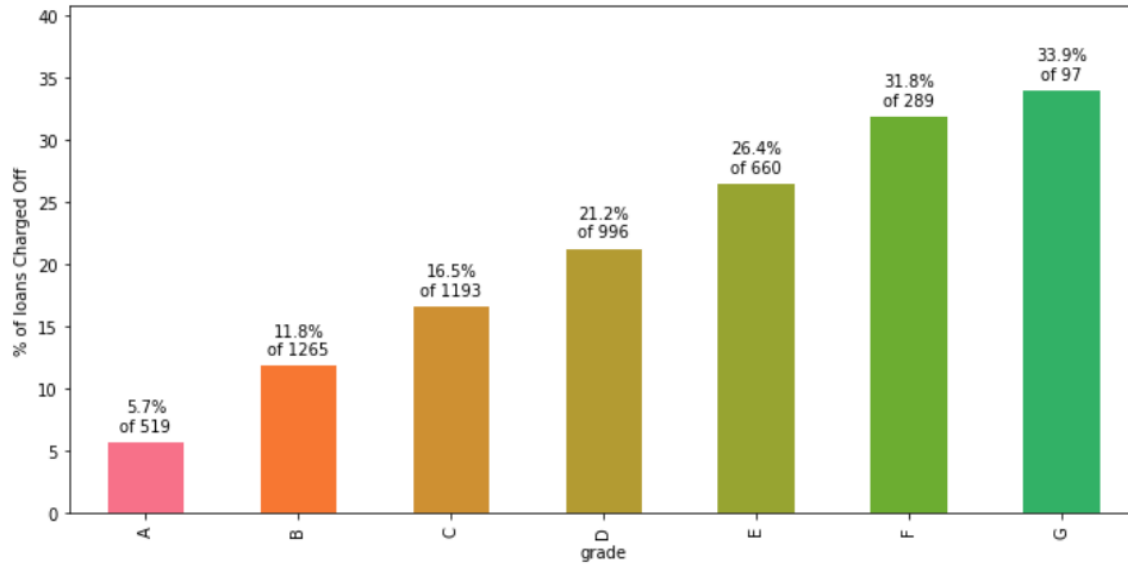


Maximum number of loans are for debt consolidation, followed by credit cards



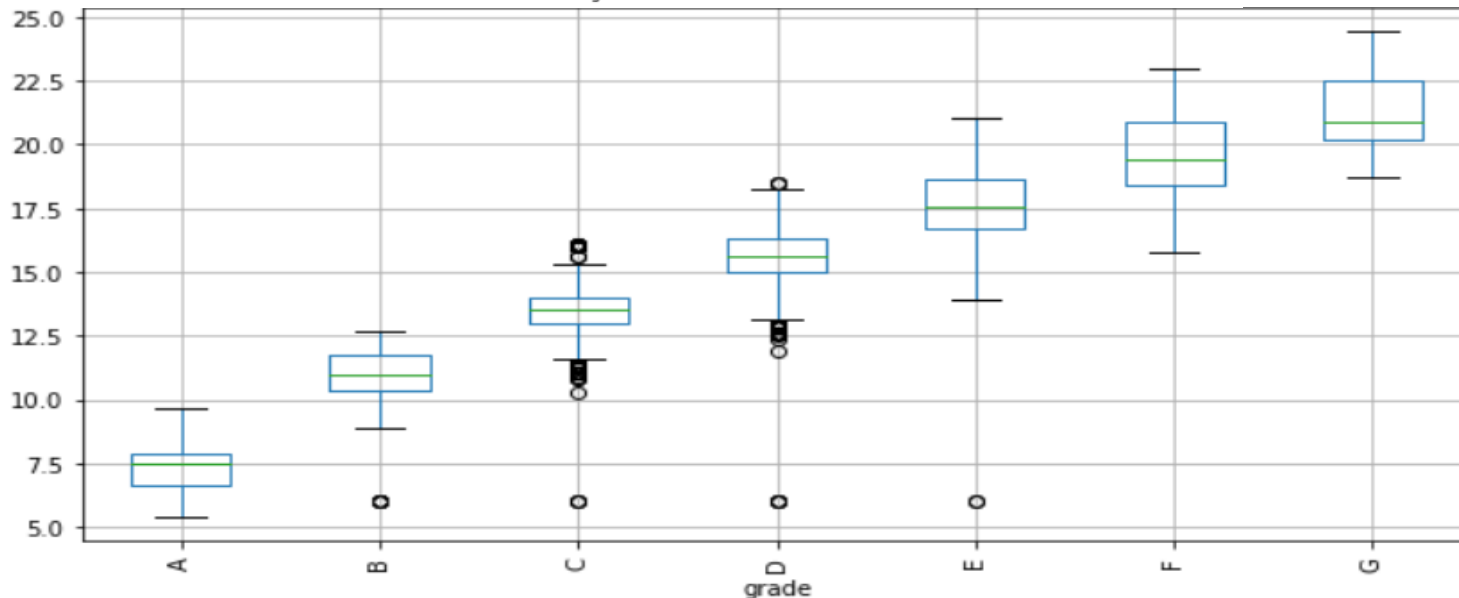
Most loans are high quality, Grade A or Grade B

Analysis-Understanding Loans Continued

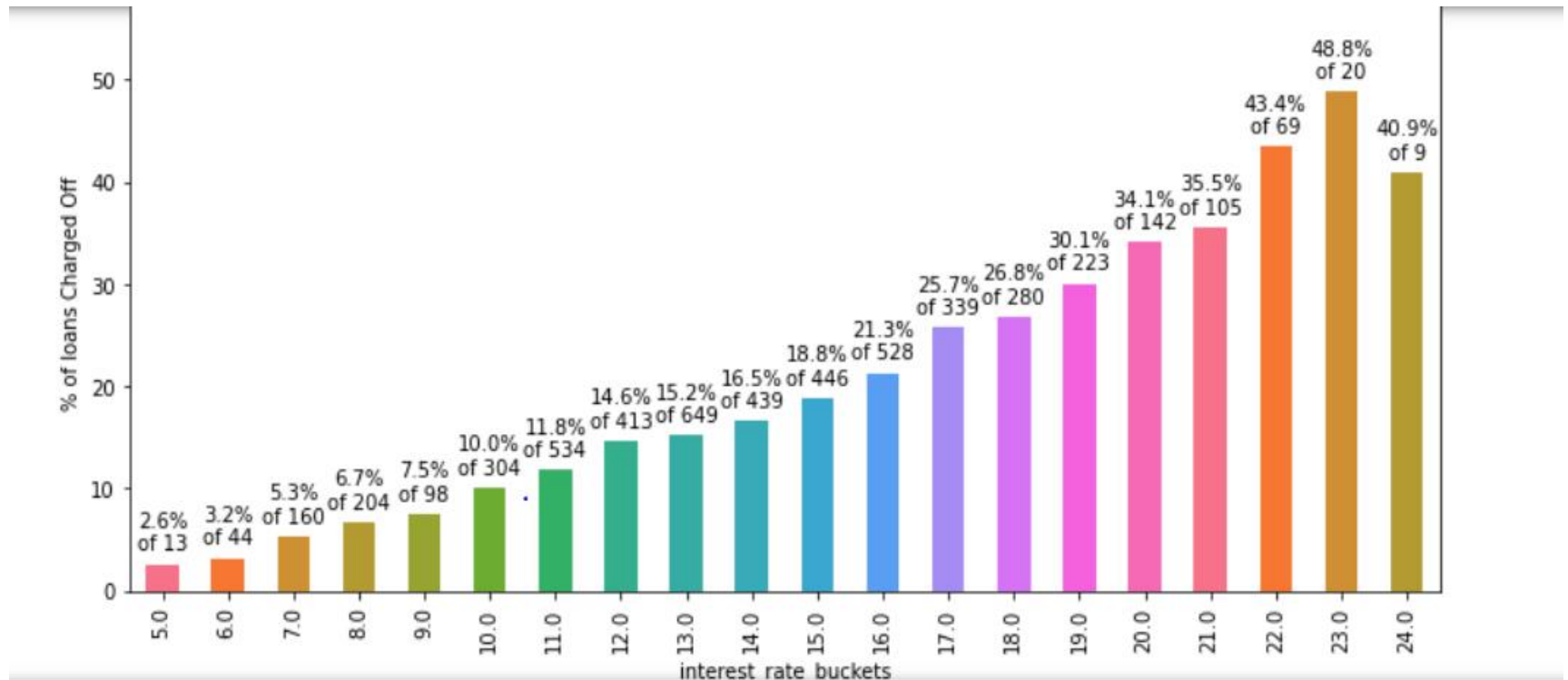


Lower grades have higher incidence of default

Higher Interest rate is charged as the grade of loan becomes worse

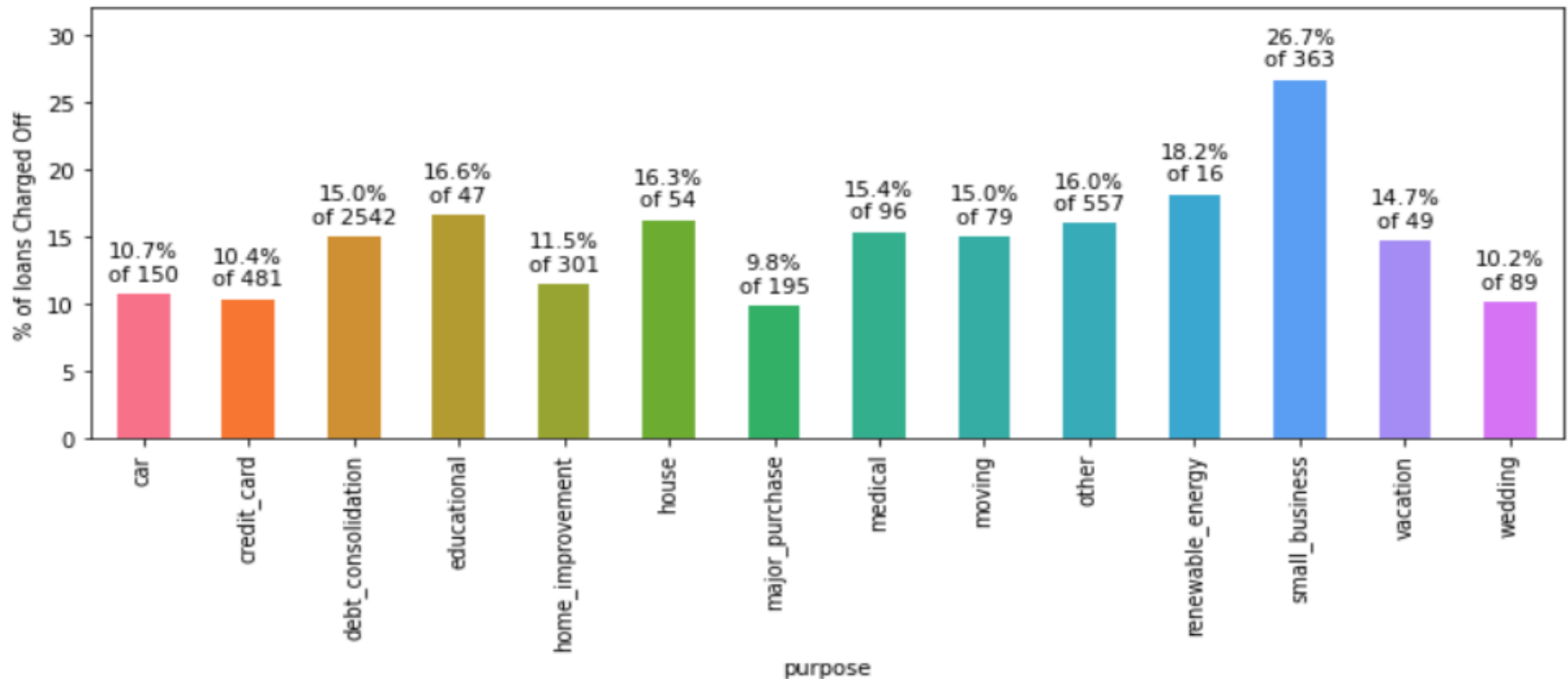


Analysis– Defaults by Interest Rates



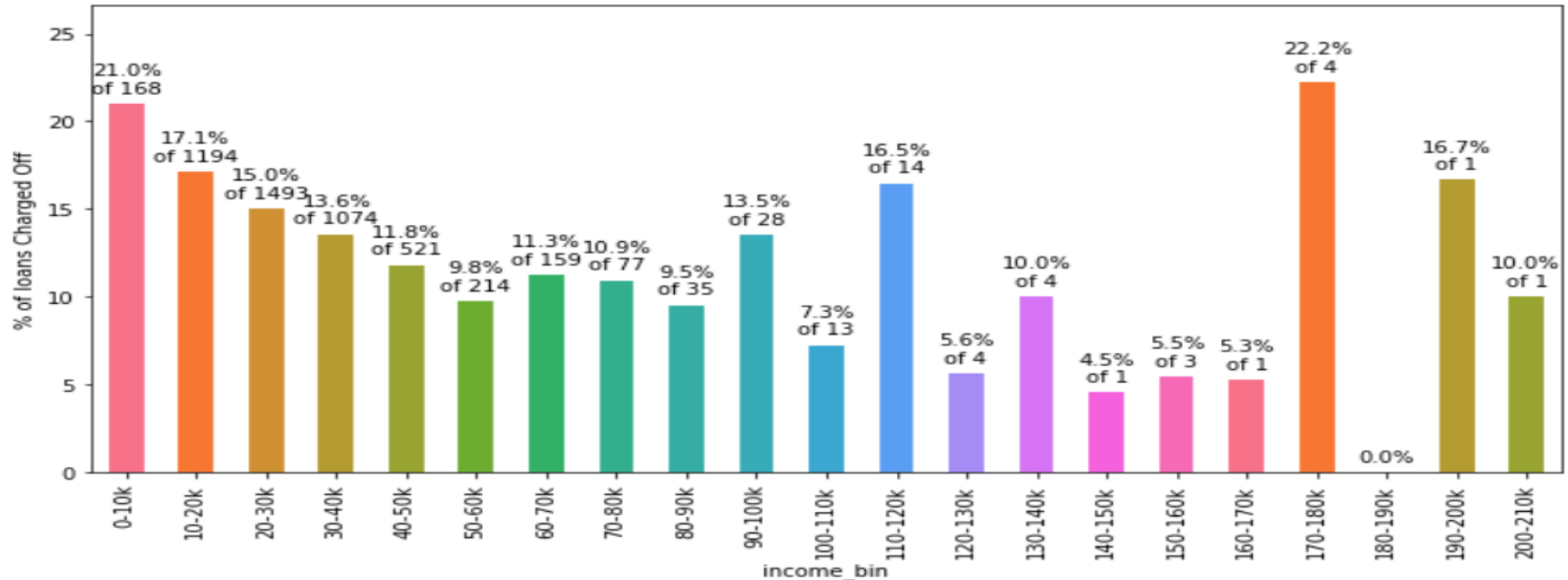
**Percentage of Defaults Increase as the rate of Interest Increases.
At rates of 20% and above more than 34% loans are Charged-Off**

Analysis–Defaults by Loan Purpose



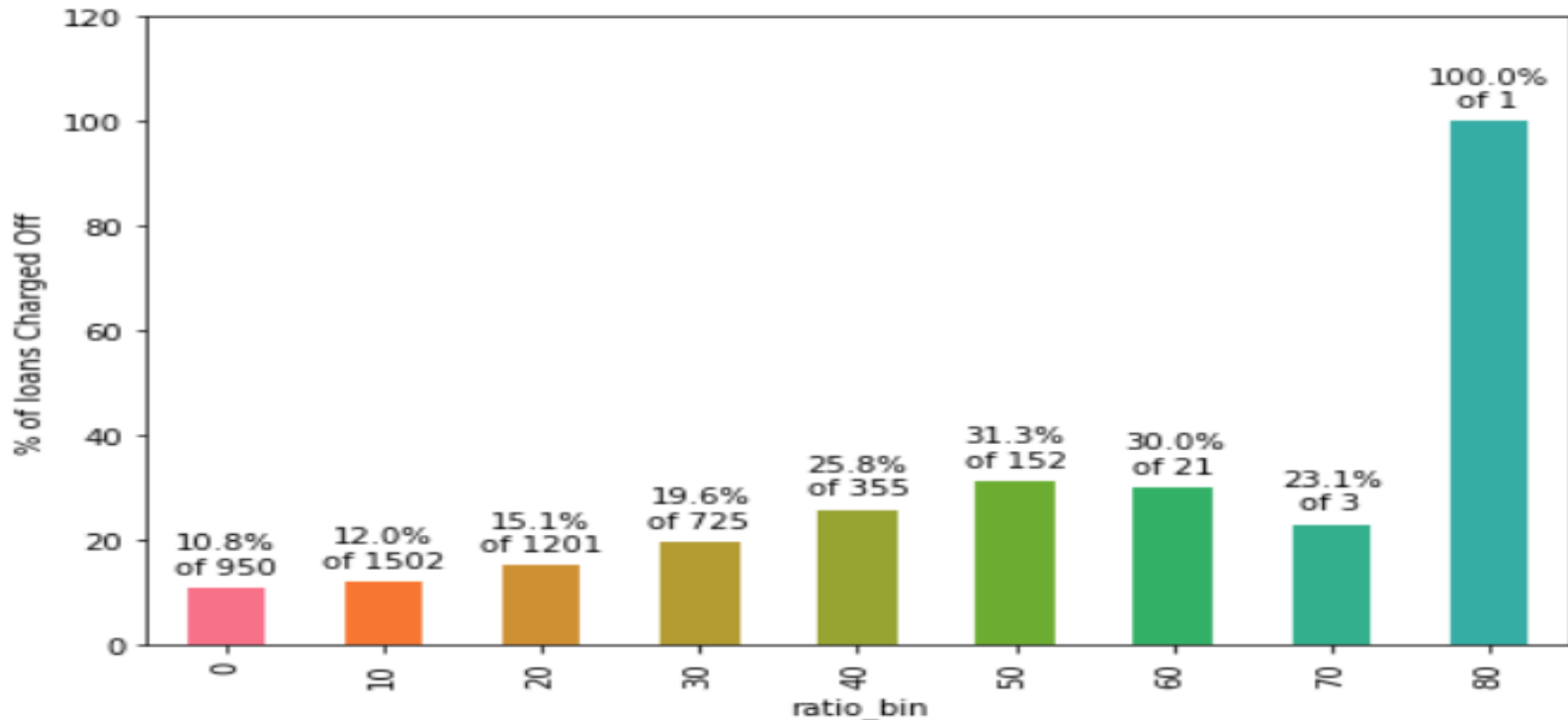
More than a quarter of the loans are taken for the purpose of running a small business

Analysis-Default's by Borrower's income



Borrower's having annual income less than 20000 default on their loans at a much higher rates, Loan Default decrease with higher annual income

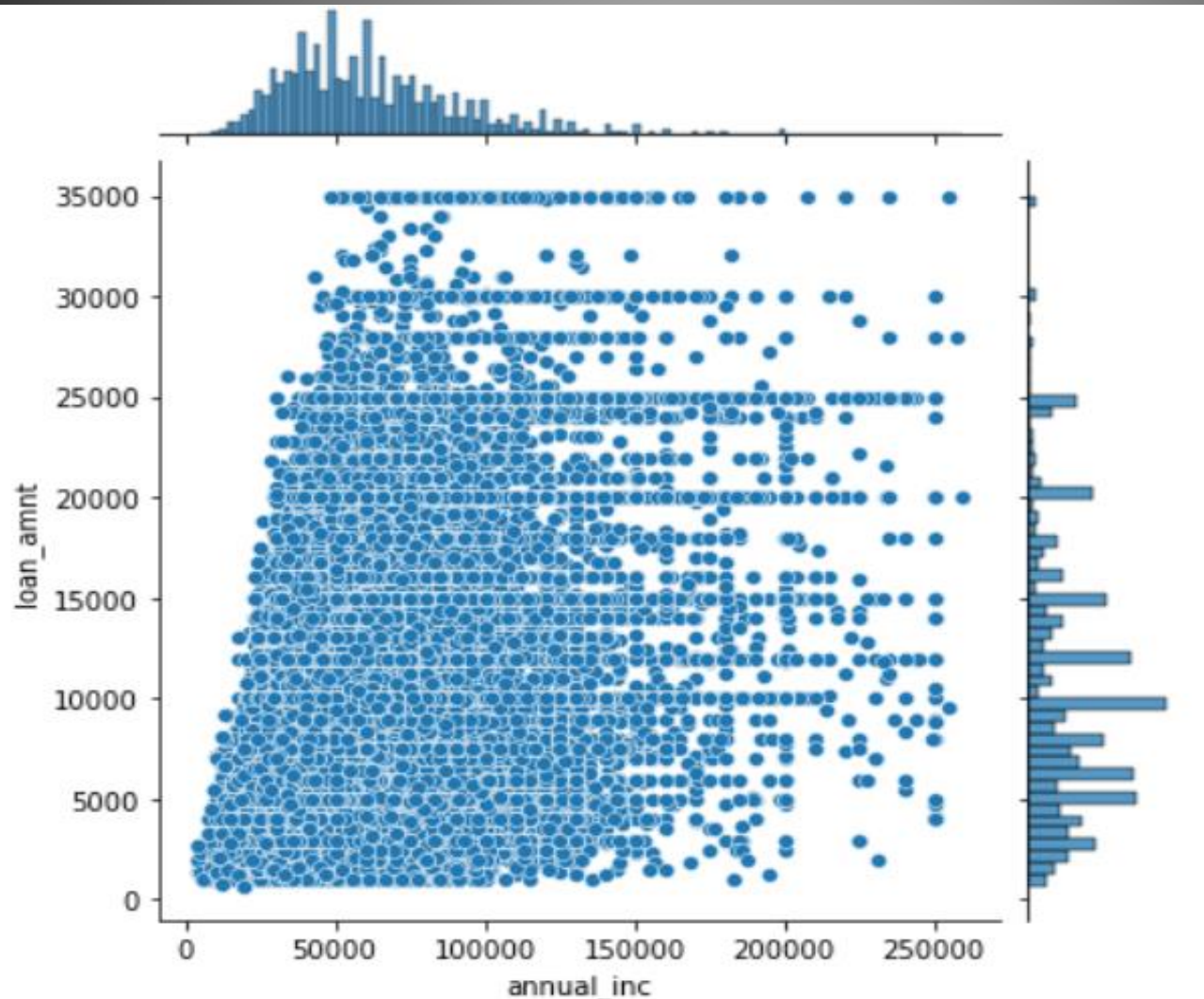
Analysis-Defaults by Ratio of amount to Income



As long as loan amount is less than 20% of the annual income defaults are less

Loan Amounts of 30 % of annual income see higher defaults

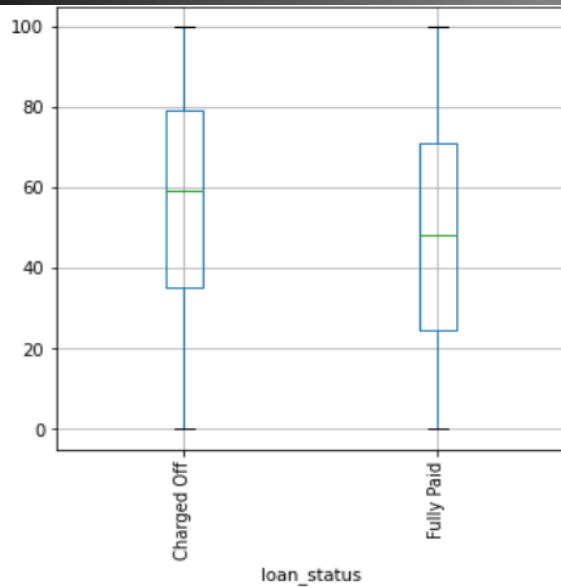
Analysis-Defaults by Ratio Continued



Lending club
has extended
high value
loans to
people with
low income

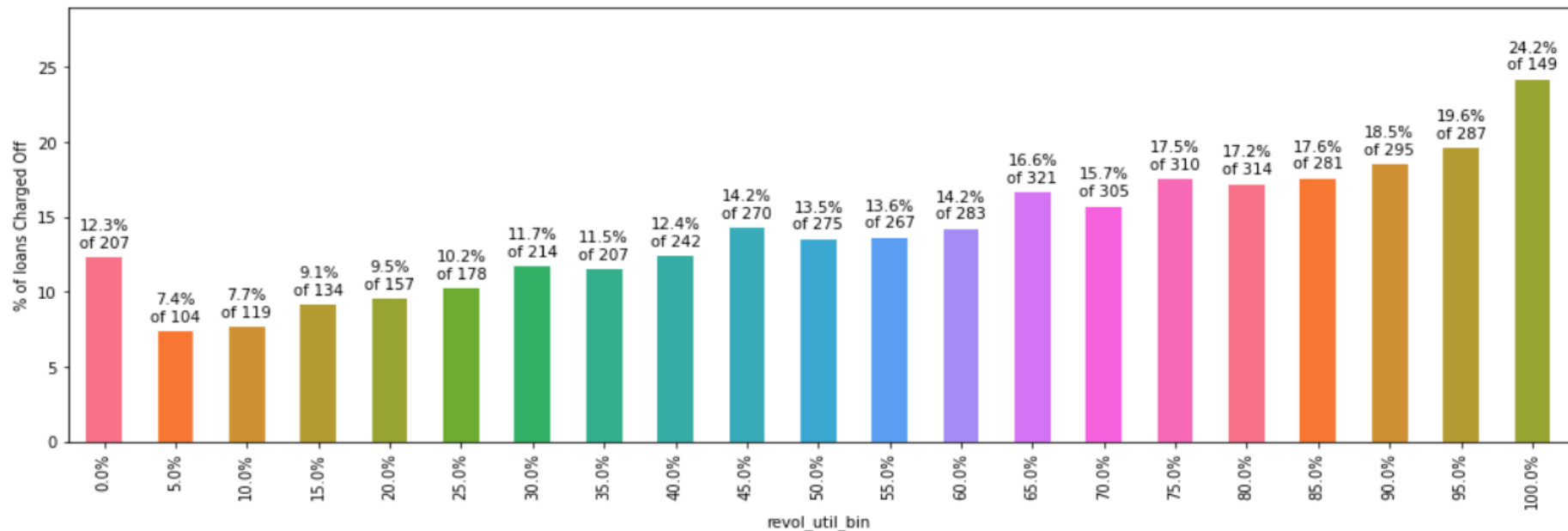
This
practice
should be
reduced

Analysis-Defaults by Revolving Line Util Rate

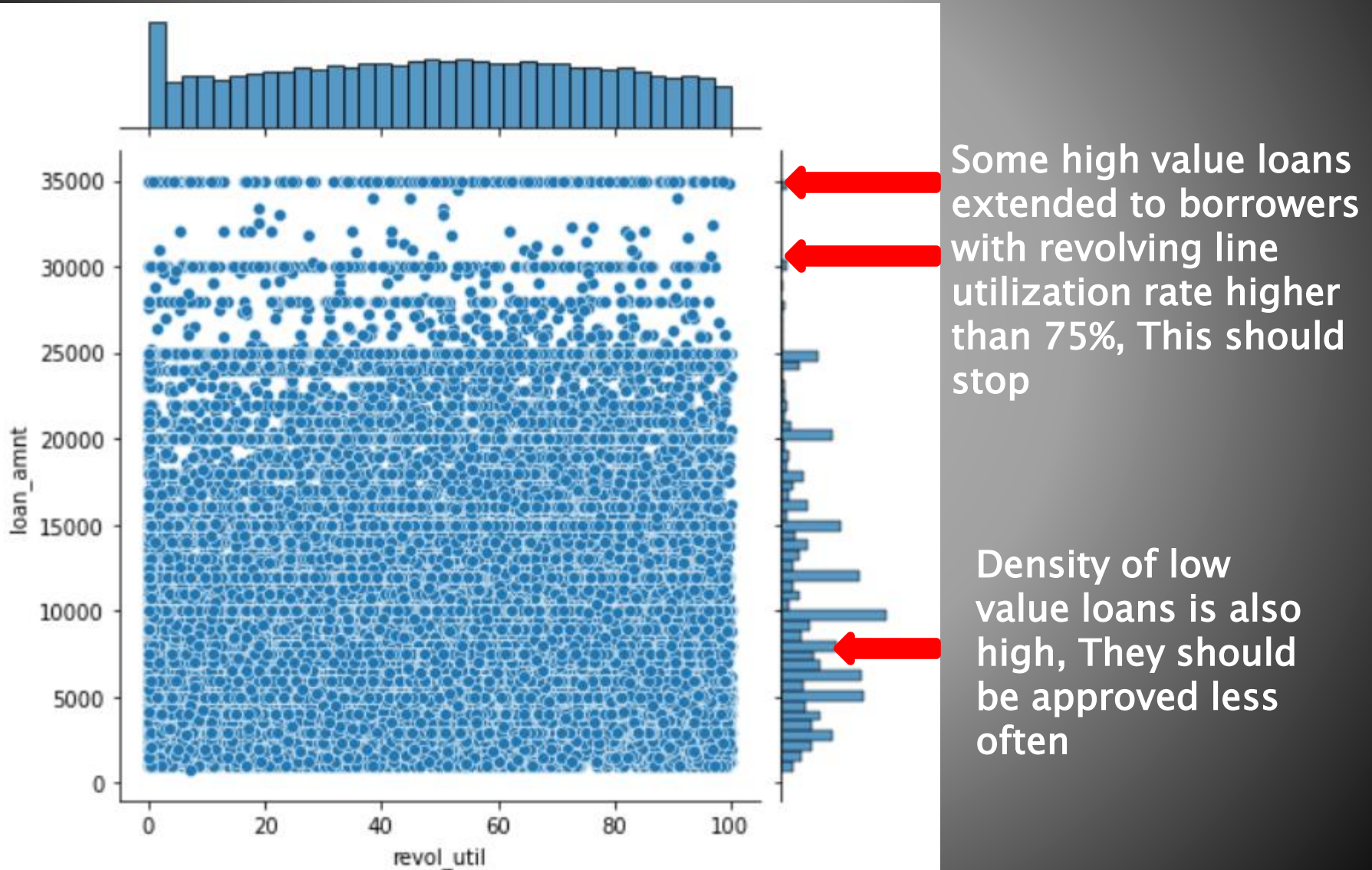


People with high utilization of Revolving Line of Credit at the time of taking loan default more

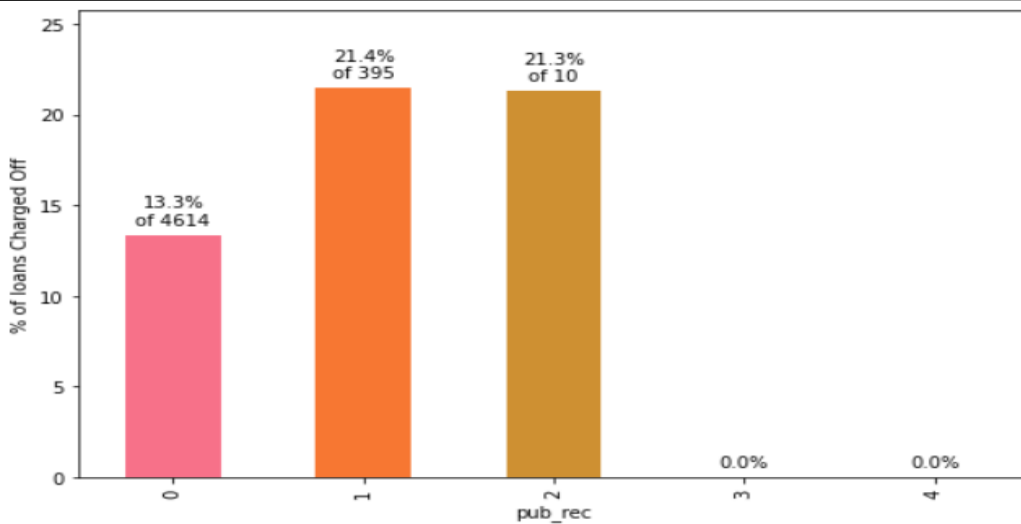
Loans with utilization > 75% are risky



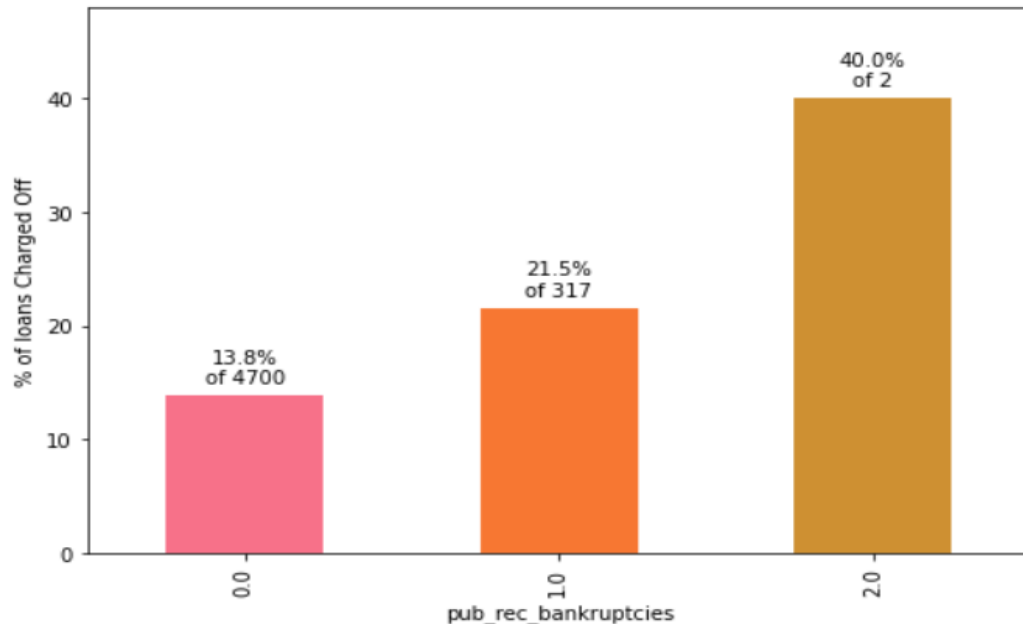
Analysis– Defaults by Rev Util Continued



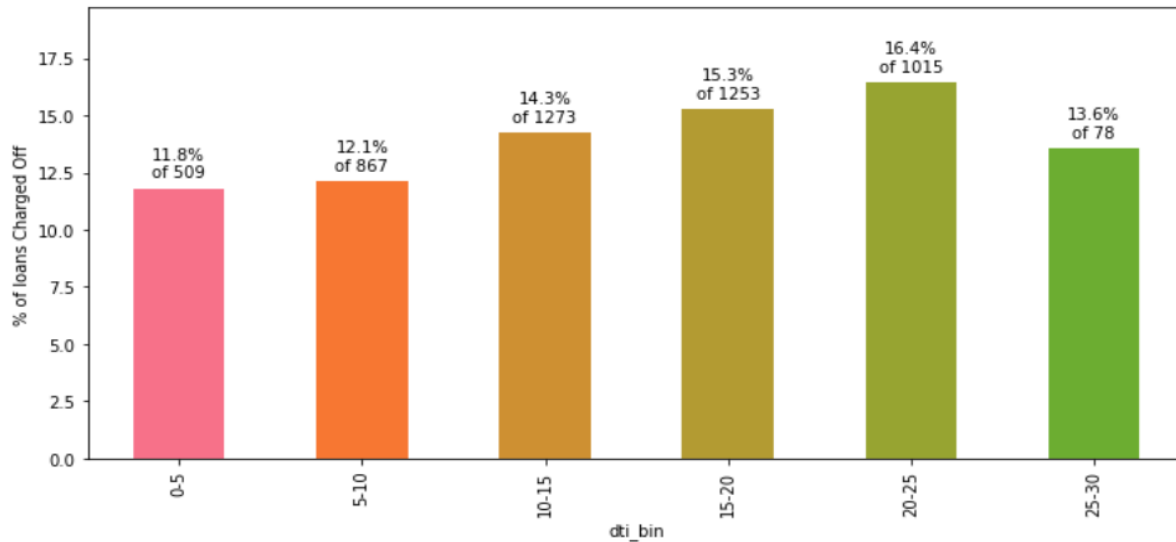
Analysis – Defaults by Prior Bad Record



Having even one derogatory or bankruptcy record increases chances of Charge-Off

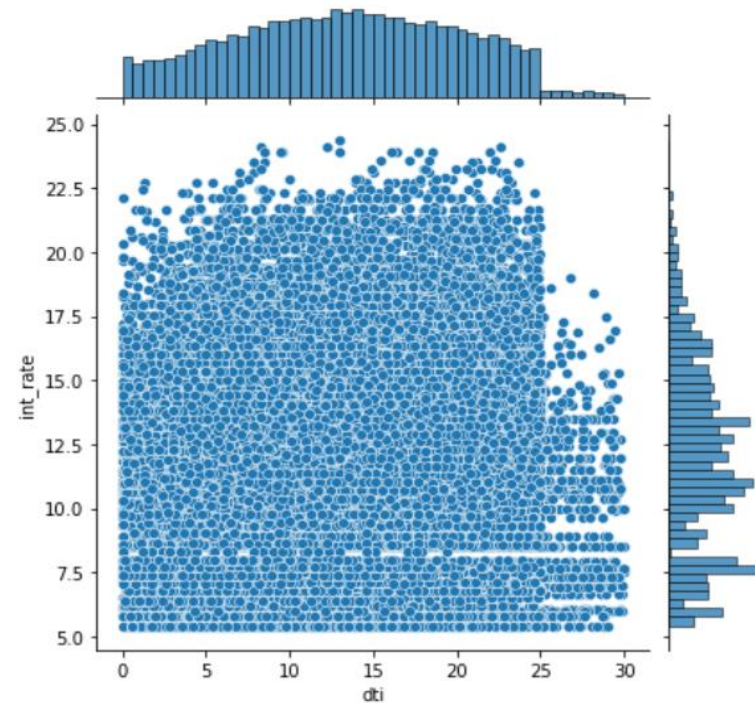


Analysis-Defaults by Debt to



Percentage of default rises with dti ratio. As the dti ratio rises above 20, the loans become risky

Higher interest rates should be charged for higher dti, but we see spread across all values



Observation Summary

- Approving loans with amounts 30% of the annual income is risky
- Approval high-value loans when revolving line utilization rate greater than 75% is not favorable
- Stop approving loans to people with prior bad record
- Start charging higher interest rates for loans with dti greater than 20

Thank You